

Catalog Description

Structure of computers, past and present: first, second, third and fourth generation. Combinatorial and sequential circuits. Programmable logic arrays. Processor design: information formats, instruction formats, arithmetic operations and parallel processing. Hardwired and microprogrammed control units. Virtual, sequential and cache memories. Input-output systems, communication and bus control. Multiple CPU systems.

Prerequisites

EECE.3170 Microprocessor Systems Design I; EECE.2650 Introduction to Logic Design

Course Learning Outcomes

Upon completion of this course, the students will demonstrate the ability to:

1. Design, test, and evaluate multiple CPU implementation styles
2. Describe microarchitectural elements in a computing system
3. Model microarchitectural elements in a computing system
4. Collect relevant benchmarking data for microarchitectural elements and evaluate their performance

Instructor

Orlando R. Arias, PhD
Assistant Professor
orlando_arias@uml.edu
Ball 407A

Office Hours

TBD.

Office hours may be subject to cancellation. In such event, I will attempt to notify all students ahead of time and reschedule office hours during that week.

Teaching Assistant

TBD.

Course Sections

See SiS for details. Link to virtual course will be provided before the start of the semester.

Textbook

No text book is required for this class. We will utilize academic and industry papers as needed to cover materials.

Additional Resources

In an attempt to encourage communications and collaboration, we will utilize the discussion forums on Canvas as well as a dedicated Slack Workspace. The Slack Workspace will be linked to in Canvas.

Other resources will be placed or linked to on Canvas.

Canvas and Echo 360

Course notes, projects, and homeworks will be available on Canvas. Submissions will be done through Canvas. On projects, students will need to submit proof their design works on hardware as part of their report.

Course Structure & Schedule

This course will be taught virtually.

Attendance and Participation

While not strictly required, attendance to lectures is highly encouraged. Students are encouraged to ask questions during class: *there are no wrong questions*. Material covered in class will play an integral role in student success.

Software & Hardware Requirements

Assignments for this class will make use of different tools. The use of a GNU/Linux environment is highly desired. Windows users may use WSL2 with up-to-date software. A C++ compiler that supports the 2020 revision of the standard will be necessary for some assignments. Modern versions of GCC and LLVM/Clang support this standard. A modern Python distribution will be required. Extra Python packages will be given on a per-project basis. A full RV64 cross compiler toolchain will be needed for some assignments.

Graduate students will be required to use the Intel Pin dynamic instrumentation framework or similar tool for extra data collection. Students will *not* be tasked with creating their own dynamic instrumentation tools from scratch.

The Fedora computers in Ball Hall 422 have all the necessary tooling for this class. Furthermore, instructor will provide a Docker container which will have all the necessary tools preinstalled.

Grading

Students will be graded based on the completeness of their work. Homeworks are to be done individually, but collaboration among students is encouraged. Please report any collaborative efforts on solving homework problems to avoid the system from automatically flagging plagiarism. *Students will not be penalized for collaborating as long as the collaboration is reported.*

Every project requires a working demonstration on the Terasic DE10-Lite board as well as a project report. Each project will have additional submission instructions.

Tentative Grading

Homeworks are worth 15% of your final grade.

Project	Summary	
Multicycle datapath	Multicycle datapath modeling and control units	20%
Pipelining	Pipeline design and data hazards	20%
Branch accelerators	Branch predictors, branch target buffers, shadow stacks	20%
Out-of-order Execution	In-order fetches, out-of-order dispatch, in-order commits	25%

There are no exams for this class.

Late Work Policy

Please try to complete your work in a timely manner. Late assignments will be accepted, however points will be deducted. Point deduction will be voided under some exceptions (medical, emergencies...) at the discretion of the instructor.

Late Work Penalties

1 day late 10% penalty

2 to 6 days late 20% penalty

7 or more days late 30% penalty

Please have all your assignments submitted by August 12, 2026. Failure to do so will result in a grade of 0 for any missing assignments.

Grading Scale

Undergraduate		Graduate	
Score	Range Grade	Score	Range Grade
96 – 100+	A	≥ 100	A+
92 – < 96	A-	96 – < 100	A
88 – < 92	B+	92 – < 96	A-
84 – < 88	B	88 – < 92	B+
80 – < 84	B-	84 – < 88	B
76 – < 80	C+	80 – < 84	B-
72 – < 76	C	76 – < 80	C+
68 – < 72	C-	72 – < 76	C
64 – < 68	D+	< 72	F
60 – < 64	D		
< 60	F		

Depending on grade distribution on the undergraduate class, I may curve the grades. If I curve grades, I will *not* decrease grades, nor will I modify the grading scale to fit a normal distribution. That is to say, grades will be curved so that the median falls on a C+ if and only if the median grade is below a 76.

If at any point during the class you are concerned about your grade, please see me during office hours or schedule a meeting with me.

Generative AI Policy

UMass Lowell's Academic Integrity Policy explicitly identifies unauthorized use of AI tools by students as academic dishonesty. For the purposes of this class, usage of generative AI is considered the use of an external resource. If you used generative AI in your project you must state it on your report.

Please be aware that generative AI can “hallucinate”. If you choose to use generative AI, you are fully responsible for ensuring the correctness of the output.

Course Topics

The following is the tentative list of topics that will be covered in class.

- **Introduction** Introduction and class expectations.
- **Instruction Set Architectures** Types of ISA, ISA designs and considerations. The AMD64 and RISC-V ISA.
- **ALU and Datapath Designs** Multicycle datapaths, control units, introduction to microcode.
- **Pipelined Designs** Pipelined microarchitectures, data hazards and forwarding, interlocking, branch resolution and branch prediction.
- **Memory Hierarchies** Memory and cache hierarchies. Address translation.
- **Out-of-Order Execution** Out-of-order designs. Preserving exception state, in order commits.
- **Microarchitectural Security** Side channels and covert channels.

Disability Statement

Any student who feels they need an accommodation based on the impact of a disability should contact the instructor privately to discuss specific needs. Please contact the UML Office for Disability Services (<https://www.uml.edu/student-services/disability/>) for assistance in verifying the need for accommodations and developing accommodation strategies.

Please contact me with any requests as soon as possible.

Academic Misconduct Statement

Any student found to have engaged in academic misconduct, as set forth in the Code of Student Conduct <https://www.uml.edu/Catalog/Undergraduate/Policies/Academic-Policies/Academic-Integrity.aspx>, will be subject to disciplinary action by the university. Academic misconduct is any activity that tends to compromise the academic integrity of the university, or subvert the education process.

Student Conduct

Students are expected to abide by the provisions in the Code of Student Conduct. The University's Code of Student Conduct, <https://www.uml.edu/student-services/Student-Conduct/Student-Conduct-Code.aspx>, and Sexual Misconduct Policy, <https://www.uml.edu/student-services/Student-Conduct/sexual-misconduct-policy.aspx>

University Privacy Statement

UMass Lowell recognizes the importance of mutual trust between students and faculty. Neither faculty nor students may record video or audio of a course or private conversation without all parties' consent. Massachusetts is a two-party consent state, which means it is illegal to record someone without their permission. Recordings of classroom lectures are the intellectual property of the instructor. Instructors have the right to prohibit audio and video recording of their lectures, unless the requesting student is registered with Disabilities Services and recording of class sessions is an approved accommodation. In addition, sharing of or selling recordings of classroom activity, discussions or lectures with any other person or medium without permission of the instructor is prohibited.

Tutoring Resources

The Centers for Learning and Academic Support Services (CLASS) provide tutoring services, including an online searchable tutoring schedules are available that include resources on all campuses. A tutoring request form is also available if there are no tutors listed for the class for which you need help.

Student Mental Health and Wellbeing

We are a campus that cares about the mental health and well-being of all individuals in our campus community. Your personal health and well-being can impact your success in this course. Students sometimes experience mental health concerns or stressful experiences that interfere with academics and have a negative impact on everyday life. If you or someone you know are experiencing mental health challenges at UMass Lowell, please contact Counseling (information below). Their services are free and confidential, and same day appointments are available.

I am available to talk with you about stresses related to your work in my class. Furthermore, I can assist you in reaching out to any one of a wide range of campus resources, including:

Counseling Services provides crisis intervention, assessment, referrals, short-term individual counseling, group therapy, and on-call clinicians outside of business hours. They are located at University Crossing Suite 300 and their 24/7 phone number is 978-934-6800.

UMatter2 is a university-wide initiative to support students and promote mental health. The office may be reached at 978-934-6671.

Consider also reaching out to a friend, faculty or family member you trust for help getting connected to the support that can help.